

Razorpay Integration Audit

Shree Shyam Dairy Farm · June 27, 2026

Next.js 16 · TypeScript · Vercel · Neon PostgreSQL

Overall completion: ~82% — Steps 1–6 implemented and deployed on production.

Executive Summary

Razorpay integration Steps 1–6 are implemented and deployed. Create-order, verify, checkout UI, payment persistence, inventory, notifications, and webhook routes all exist. Main gaps: no confirmed live payment in DB yet, webhook Dashboard registration unverified, public /test-payment page should be removed after testing.

1. Environment Variables

Variable	Status	Usage
NEXT_PUBLIC_RAZORPAY_KEY_ID	Configured	Public key, checkout popup
RAZORPAY_KEY_SECRET	Configured	Server-only, verify & create-order
RAZORPAY_WEBHOOK_SECRET	Configured	Webhook signature verification
		• Production /api/payment/config returns configured: true, live key rzp_live_... • Secret never exposed in API responses • Zod validation in src/config/payment.ts

2. Razorpay SDK

- Package: razorpay@^2.9.6 in package.json
- Client: src/lib/razorpay.ts with server-only guard
- Singleton with HMR-safe caching

3. Create Order API

Path: src/app/api/payment/create-order/route.ts

- Validates amount > 0, currency, receipt
- INR major unit to paise conversion
- Production tested — returns Razorpay order ID
- Links internal DB order to razorpayOrderId

4. Frontend Checkout

Component	Path	Status
Cart modal	PaymentCheckoutModal.jsx	Active
Reusable	RazorpayCheckout.tsx	Active

Test page

/test-payment

Deployed

Flow: checkout !' config !' create-order !'
checkout.js !' popup !' verify

5. Payment Verification

Path: src/app/api/payment/verify/
route.ts

- HMAC SHA-256 with timingSafeEqual
- Persists payment, marks order PAID
- Inventory, invoice, notifications
- Idempotent duplicate payment handling

6. Webhook

Path: src/app/api/payment/webhook/
route.ts

- Raw body + x-razorpay-signature verification
- Events: payment.*, order.paid, refund.*
- Production: GET !' 405, POST without sig !' 401 (route exists)
- 404 was from old deployment before route existed

Register URL: <https://shree-shyam-dairy-farm.vercel.app/api/payment/webhook>

Completed

- All 3 env vars on Vercel
- SDK singleton (src/lib/razorpay.ts)
- Create Order API (TS, validated, prod-tested)
- Frontend checkout (modal, component, test page)
- Verification (HMAC + persistence + inventory)
- Webhook (signature, events, idempotency)
- Database (Neon, 5 migrations)
- Production deploy on main branch

Partially Completed

- Legacy JS routes (config, upi-qr, qr-status) not migrated
- Duplicate SDK client (razorpayServer.js)
- Dual checkout paths (tsx vs js utility)
- 0 Payment rows in DB — no live payment confirmed
- No payment route integration tests
- No rate limiting on payment APIs

Missing

- Payment E2E / integration tests
- Razorpay Dashboard webhook registration confirmed
- Customer order history UI
- Rate limiting on /api/payment/*
- CI gate on deploy
- Remove /test-payment after testing

Bugs

Severity

Medium
Low
Low

Severity

High
Medium
Medium
Low

Dimension

Environment

Issue

No completed payments in DB despite pending orders
Dual checkout implementations (drift risk)
Client estimatedTotal trusted for order total

Security Issues

Issue

Public /test-payment allows live 11 charges
No rate limit on create-order
Client-supplied amount in verify
Legacy razorpayServer.js without server-only

Deployment Issues

- Webhook 404 was stale deploy — now live (401/405)
- Register webhook URL + secret in Razorpay Dashboard
- CI tests red; deploy not gated

Production Readiness Scores

Score /10

8.5

SDK	8.5
Create Order API	9.0
Checkout	8.5
Verification	9.0
Webhook	7.5
Deployment	8.0
Security	7.0
Code Quality	7.5
Overall	~82%

Next Steps (P0 Critical)

- Complete ¹1 live payment test via /test-payment
- Register webhook in Razorpay Dashboard
- Verify webhook delivery returns 200
- Remove or protect /test-payment page

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Read-only audit, no code modified.